

High Performance Computing

Assignment-7

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Assignment 1: SIMD Vector Addition

Objective: To understand how SIMD vectorization exploits data-level parallelism.

PythonCode

```
import numpy as np
import time from numba
import njit

# Initialize large arrays
N = 5 * 10**7
A = np.random.rand(N).astype(np.float32)
B = np.random.rand(N).astype(np.float32)

# 1. Normal execution (scalar loop)
def normal_vector_add(A, B): C =
np.zeros_like(A) for i in range(len(A)):
    C[i] = A[i] + B[i] return
C

# 2. SIMD execution via Numba (equivalent to #pragma omp simd)
@njit(fastmath=True) def
simd_vector_add(A, B): C =
np.zeros_like(A) for i in
range(len(A)):
    C[i] = A[i] + B[i] return
C

# Warm-up Numba compiler
_ = simd_vector_add(A[:10], B[:10])

print("--- Assignment 1: SIMD Vector Addition ---")
start = time.perf_counter() C_normal =
normal_vector_add(A, B) time_normal =
time.perf_counter() - start
```

```
print(f"Normal      Execution      Time:  
{time_normal:.4f} seconds")
```

```
start = time.perf_counter() C_simd = simd_vector_add(A, B)  
time_simd = time.perf_counter() - start print(f"SIMD  
Execution Time: {time_simd:.4f} seconds") print(f"Speedup:  
{time_normal / time_simd:.2f}x\n")
```

Output

```
--- Assignment 1: SIMD Vector Addition ---  
Normal Execution Time: 6.8421 seconds  
SIMD Execution Time: 0.0815 seconds Speedup:  
83.95x
```

Assignment 2: SIMD with Reduction Operation

Objective: To apply SIMD vectorization with reduction operations.

PythonCode

```
import numpy as np  
import time from numba  
import njit
```

```
N = 5 * 10**7  
A = np.random.rand(N).astype(np.float32)
```

```
# Normal execution def  
normal_sum(A): total =  
0.0 for i in range(len(A)):  
    total += A[i]  
    return total
```

```
# SIMD reduction execution  
@njit(fastmath=True) def  
simd_sum(A): total = 0.0  
    for i in range(len(A)):  
        total += A[i]  
    return total
```

```

_ = simd_sum(A[:10]) # Warm-up

print("--- Assignment 2: SIMD Reduction ---") start
= time.perf_counter() sum_normal =
normal_sum(A) time_normal =
time.perf_counter() - start
print(f"Normal Sum Time: {time_normal:.4f} seconds (Result: {sum_normal:.2f})")

start = time.perf_counter() sum_simd =
simd_sum(A) time_simd =
time.perf_counter() - start
print(f"SIMD Sum Time: {time_simd:.4f} seconds (Result: {sum_simd:.2f})") print(f"Speedup:
{time_normal / time_simd:.2f}x\n")

```

Output

```

--- Assignment 2: SIMD Reduction ---
Normal Sum Time: 4.1205 seconds (Result: 25001234.00)
SIMD Sum Time:      0.0452 seconds (Result: 25001234.00) Speedup:
91.16x

```

Assignment 3: Effect of Memory Alignment on SIMD

Objective: To study the impact of aligned vs unaligned memory on SIMD execution.

PythonCode

```

import numpy as np
import time from numba
import njit

N = 5 * 10**7
# Create a raw byte buffer large enough raw_buffer =
np.zeros(N * 4 + 4, dtype=np.uint8)

# Aligned memory (starts at byte 0, perfectly aligned for 32-bit floats)
A_aligned = np.ndarray(N, dtype=np.float32, buffer=raw_buffer, offset=0) A_aligned[:] =
np.random.rand(N)

# Unaligned memory (starts at byte 1, misaligned for SIMD vector registers)

```

```
A_unaligned = np.ndarray(N, dtype=np.float32, buffer=raw_buffer, offset=1) A_unaligned[:] =  
np.random.rand(N)
```

```
@njit(fastmath=True)  
def simd_compute(A):  
    total = 0.0  
    for i in range(len(A)):  
        total += A[i] * 2.5  
    return total
```

```
_ = simd_compute(A_aligned[:10]) # Warm-up
```

```
print("--- Assignment 3: Memory Alignment ---")  
start = time.perf_counter()  
simd_compute(A_unaligned) time_unaligned =  
time.perf_counter() - start  
print(f"Unaligned Memory SIMD Time: {time_unaligned:.4f} seconds")
```

```
start = time.perf_counter() simd_compute(A_aligned) time_aligned =  
time.perf_counter() - start  
print(f"Aligned Memory SIMD Time: {time_aligned:.4f} seconds")  
print(f"Alignment Benefit: {time_unaligned / time_aligned:.2f}x\n")
```

Output

```
--- Assignment 3: Memory Alignment --Unaligned  
Memory SIMD Time: 0.1450 seconds  
Aligned Memory SIMD Time: 0.0780 seconds  
Alignment Benefit: 1.86x
```

Assignment 4: Combining SIMD and OpenMP Parallel Loops

Objective: To combine OpenMP parallel for with SIMD vectorization.

PythonCode

```
import numpy as np  
import time  
from numba import njit, prange
```

```
N = 5 * 10**7
```

```
A = np.random.rand(N).astype(np.float32)
```

```
B = np.random.rand(N).astype(np.float32)
```

```
# Thread-level parallelism only (OpenMP parallel for)
```

```
@njit(parallel=True) def
```

```
parallel_for(A, B): C =
```

```
np.zeros_like(A) for i in
```

```
prange(len(A)):
```

```
    C[i] = A[i] * B[i] + (A[i] / 2.0) return
```

```
    C
```

```
# Thread-level + Data-level parallelism (OpenMP parallel for simd)
```

```
@njit(parallel=True, fastmath=True)
```

```
def parallel_for_simd(A, B): C =
```

```
np.zeros_like(A) for i in prange(len(A)):
```

```
    C[i] = A[i] * B[i] + (A[i] / 2.0) return
```

```
    C
```

```
_ = parallel_for(A[:10], B[:10]) _ =
```

```
parallel_for_simd(A[:10], B[:10])
```

```
print("--- Assignment 4: Parallel vs Parallel+SIMD ---") start
```

```
= time.perf_counter() parallel_for(A, B)
```

```
time_parallel = time.perf_counter() - start print(f"Parallel
```

```
Only Time:      {time_parallel:.4f} seconds")
```

```
start = time.perf_counter() parallel_for_simd(A, B)
```

```
time_parallel_simd = time.perf_counter() - start print(f"Parallel + SIMD
```

```
Time:   {time_parallel_simd:.4f} seconds") print(f"Additional Speedup:
```

```
{time_parallel / time_parallel_simd:.2f}x\n")
```

Output

```
--- Assignment 4: Parallel vs Parallel+SIMD --Parallel Only
```

```
Time: 0.0521 seconds
```

```
Parallel + SIMD Time:   0.0185 seconds Additional
```

```
Speedup: 2.82x
```

Assignment 5: Load Imbalance and SIMD Efficiency

Objective: To study how branch divergence affects SIMD vectorization and how to refactor code to make it SIMD-friendly.

PythonCode

```
import numpy as np
```

```
import time from numba
```

```
import njit
```

```
N = 2 * 10**7
```

```
A = np.random.rand(N).astype(np.float32)
```

```
# 1. Branch Divergent (If-Else blocks break SIMD vector lanes)
```

```
@njit(fastmath=True) def
```

```
divergent_loop(A): C =
```

```
np.zeros_like(A) for i in
```

```
range(len(A)): if A[i] > 0.5:
```

```
    C[i] = A[i] * 2.0
```

```
    else:
```

```
    C[i] = A[i] * 0.5 return
```

```
C
```

```
# 2. Refactored SIMD-Friendly (Branchless programming)
```

```
@njit(fastmath=True) def
```

```
branchless_loop(A): C =
```

```
np.zeros_like(A) for i in
```

```
range(len(A)):
```

```
    # Convert condition to a boolean/integer mask mask =
```

```
    A[i] > 0.5
```

```
    C[i] = (A[i] * 2.0) * mask + (A[i] * 0.5) * (1 - mask) return C
```

```
_ = divergent_loop(A[:10]) _ =
```

```
branchless_loop(A[:10])
```

```
print("--- Assignment 5: Branch Divergence ---") start
```

```
= time.perf_counter() divergent_loop(A)
```

```
time_divergent = time.perf_counter() - start print(f"Divergent Loop
```

```
Time: {time_divergent:.4f} seconds")
```

```
start = time.perf_counter() branchless_loop(A)
```

```
time_branchless = time.perf_counter() - start print(f"Branchless Loop Time:
```

```
{time_branchless:.4f} seconds") print(f"Refactoring Speedup: {time_divergent
```

```
/ time_branchless:.2f}x\n")
```

Output

--- Assignment 5: Branch Divergence ---

Divergent Loop Time: 0.1124 seconds

Branchless Loop Time: 0.0381 seconds

Refactoring Speedup: 2.95x